Dami Thomas

Cambridge, MA | 678-499-7434 | dvthomas@mit.edu | dvthomas01.github.io/Personal-Website | github.com/dvthomas01

EDUCATION

Massachusetts Institute of Technology (MIT) Cambridge, MA · June 2026
S.B. in Artificial Intelligence and Decision Making, Minor in Mechanical Engineering

- **Relevant coursework:** Introduction to Robotics, Robotic Manipulation, Introduction to Machine Learning, Advances in Computer Vision, Introduction to Autonomous Machines, Dynamics and Controls I & II, Design and Manufacturing

PUBLICATIONS

A. Thomas, R. Galliath, A. Garbuz, L. Anger, C. O'Neill, T. Johst, **D. Thomas**, G. Lordos, J. P. How. "LunarLoc: Segment-Based Global Localization on the Moon." 2025.

RESEARCH EXPERIENCE

MIT Reliable Autonomous Systems Lab (REALM) — *Undergraduate Researcher* Cambridge, MA · Aug 2025 – Present

- Trained model predictive control and PPO navigation policies for Spot robots in NVIDIA Isaac Sim industrial environments, advised by Prof. Chuchu Fan.
- Enforced real-time safety using control barrier functions for collision-free navigation.
- Deployed policies across multiple robots with independent waypoint tracking.

NASA Lunar Autonomy Challenge — *Team Member* Cambridge, MA · Oct 2024 – May 2025

- Modeled lunar surface elevation with SciPy interpolation (75–90% confidence) and built a PID controller for rover path following and boulder avoidance; contributed to the LunarLoc localization work above.
- Placed 2nd nationally among 31 U.S. university teams for autonomous rover performance.

SELECTED PROJECTS

Locomotion on Compliant Terrain — *Independent Research* https://github.com/dvthomas01/compliant-terrain-locomotion May 2026 – Present

- Trained PPO quadruped locomotion policies in MuJoCo with Stable-Baselines3, designing a controlled 2×2 ablation to test whether compliance randomization during training improves robustness on unseen compliant terrain.
- Found terrain exposure drives survival (fall rate 100% to 0%) while foot-contact history yields roughly 25% cost-of-transport efficiency gains; iterated extensively on reward shaping and training diagnostics.

Semantic Navigation Robot https://github.com/dvthomas01/semantic-robot Jul – Aug 2025

- Built a semantic visual navigation system with ROS 2 Humble, SLAM Toolbox, Nav2, YOLOv11, and Whisper, integrating perception, planning, and natural-language commands over the NVIDIA R2B dataset.

Voice-Controlled Self-Balancing Rover https://github.com/dvthomas01/voice_rover Jan 2026

- Designed a self-balancing rover (Fusion 360, Raspberry Pi, ESP32) with a 100 Hz PID balance controller and a real-time voice-to-motor pipeline using wake-word detection and Vosk speech-to-text.

INDUSTRY EXPERIENCE

National Instruments (Emerson) — *Embedded Engineering Intern* Austin, TX · June 2026 – Present

- Benchmarked gRPC communication performance between NI PXIe-8881 and NVIDIA DGX Spark hardware, evaluating grpc-direct across hardware and software configurations.

Rockwell Automation — *AI Engineer Intern* Milwaukee, WI · Jun – Aug 2025

- Built a RAG re-ranking pipeline for a pharmaceutical recipe-generation product (Streamlit, ChromaDB, LangChain, Azure OpenAI), reaching an 84% user preference rate.

Nasdaq — *AI Engineer Intern* Boston, MA · Jan – Feb 2025

- Built a year-end review synthesizer (AWS Bedrock, LLMs, LangGraph) that reduced evaluation-data collection from days to minutes via parallel processing of Jira, email, and GitHub activity.

Elinta Robotics — *Assembly and Design Engineer Intern* Kaunas, Lithuania · Jun – Aug 2023

- Designed automated assembly systems with FESTO pneumatics and built a PCB tester integrating pneumatic actuators with computer-vision quality control.

TECHNICAL SKILLS

Robotics & Control: MuJoCo, NVIDIA Isaac Sim, ROS 2, Stable-Baselines3, PPO, MPC, control barrier functions, PID control, Nav2, SLAM Toolbox
Machine Learning: PyTorch, TensorFlow, reinforcement learning, reward shaping, YOLO, CLIP, Hugging Face, NumPy, Pandas, Scikit-learn
Software: Python, MATLAB, Rust, JavaScript, C/Arduino, SQL, Git, Linux, Docker
Hardware & Maker: SolidWorks, Fusion 360, 3D printing, CNC mill and lathe, laser cutter, waterjet, welding